

TECHNICAL REPORT

NILM-Based Smart Energy Monitoring and Intelligent Consumption Alert System

Environmental Problem: High Electricity Consumption
Expected Website Solution: Smart Energy Monitoring Dashboard

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Abstract

The increasing use of electrical appliances has contributed to rising household electricity consumption. Conventional electricity meters provide the total amount of electricity consumed by a household, but they generally do not indicate how much energy is consumed by individual appliances. This makes it difficult for users to identify the major sources of excessive electricity consumption and take appropriate corrective actions.

This report proposes a NILM-Based Smart Energy Monitoring and Intelligent Consumption Alert System to address this problem. The proposed system uses Non-Intrusive Load Monitoring (NILM) to estimate the energy consumption of individual appliances from aggregate household electricity measurements. Instead of installing separate energy-monitoring devices on every appliance, the system obtains electricity measurements from a whole-house smart meter at regular time intervals. These measurements are processed by the NILM system to estimate appliance-level consumption.

The estimated data is stored and presented through a responsive web dashboard. The dashboard provides daily appliance-wise consumption, monthly comparisons, consumption trends and historical analysis. While daily data is continuously displayed, alerts are generated over a 14-day analysis period to avoid unnecessary warnings caused by normal daily fluctuations. When abnormal consumption is detected, the system identifies the appliances that contributed most to the increase and the days on which their consumption was highest.

The proposed system therefore converts raw electricity measurements into useful information that can help users understand their energy behaviour, identify inefficient consumption and make informed decisions to reduce unnecessary electricity use.

Introduction

Electricity consumption in households is increasing with the growing use of appliances such as fans, lights, refrigerators, televisions and air conditioners. Although conventional electricity meters can measure the total electricity consumed, they do not clearly show which individual appliances are responsible for that consumption. This makes it difficult for users to identify excessive energy usage and take appropriate action.

The proposed solution is a **Smart Energy Monitoring Dashboard** that provides a clear and interactive view of household electricity consumption. The dashboard displays appliance-wise daily and monthly energy usage, consumption trends, historical comparisons and important consumption patterns in an easy-to-understand form. To obtain appliance-level information without installing a separate monitoring device on every appliance, the system uses **Non-Intrusive Load Monitoring (NILM)**. Aggregate electricity measurements from a whole-house smart meter are processed by NILM to estimate the energy consumption of individual appliances.

The dashboard also analyses consumption over **14-day periods** and generates alerts when usage is significantly higher than the expected level. It can identify the appliances mainly responsible for the increase and the days on which their consumption was highest.

Thus, the main purpose of the system is to transform electricity data into useful visual information and actionable insights, helping users understand and manage their household energy consumption more effectively.

Problem Statement

Environmental Problem

The selected environmental problem is **High Electricity Consumption**.

Households may consume more electricity than necessary because of:

- prolonged operation of appliances,
- inefficient appliances,
- unnecessary lighting,
- excessive cooling,
- appliances being left operating when not required,
- lack of awareness of appliance-level energy consumption,
- inability to identify which appliance is responsible for an increase in total consumption.

The main difficulty is not simply measuring electricity consumption. Modern electricity meters can already measure overall consumption. The greater challenge is understanding where the electricity is being consumed.

Existing Problem

For example, if household consumption increases from 250 kWh to 330 kWh, the user knows that consumption increased by:

$$330 - 250 = 80 \text{ kWh}$$

However, they may not know whether the increase was caused mainly by the AC, refrigerator, fans, lighting or other appliances.

Manual monitoring is time-consuming, while installing separate monitors for every appliance can be costly.

Problem Definition

How can household electricity consumption be automatically estimated at the appliance level and presented through a centralized dashboard without installing separate monitoring devices for every appliance?

Sustainability Analysis

The proposed NILM-based energy monitoring system contributes to sustainable electricity consumption by helping users understand and control their energy usage.

Sustainable Development Goals

The system mainly supports **SDG 7 – Affordable and Clean Energy**, by promoting efficient use of electricity and helping users identify appliances with high energy consumption. It also supports **SDG 12 – Responsible Consumption and Production**, as users can identify unnecessary energy consumption and take corrective actions.

ISO 14001 Relevance

The system follows ISO 14001 principles by monitoring energy use, identifying abnormal consumption and supporting corrective action.

Life Cycle Assessment

A simplified Life Cycle Assessment (LCA) can be represented as:

Measurement → Data Analysis → Identify Excess Usage → Corrective Action → Reduced Energy Consumption

NILM can reduce the need for installing separate energy-monitoring devices on every appliance. However, the environmental impact should also be considered. A formal **EIA** is generally not required for this small-scale IT solution. Its environmental impact can be evaluated through energy savings and reduced unnecessary consumption.

Overall Impact

The system promotes energy awareness, responsible consumption and improved energy efficiency by identifying appliances contributing to excessive usage.

Proposed Engineering/IT Solution

The proposed solution is a NILM-Based Smart Energy Monitoring and Intelligent Alert System designed to monitor household electricity consumption and estimate the energy used by individual appliances without installing a separate energy-monitoring device on every appliance.

Objectives

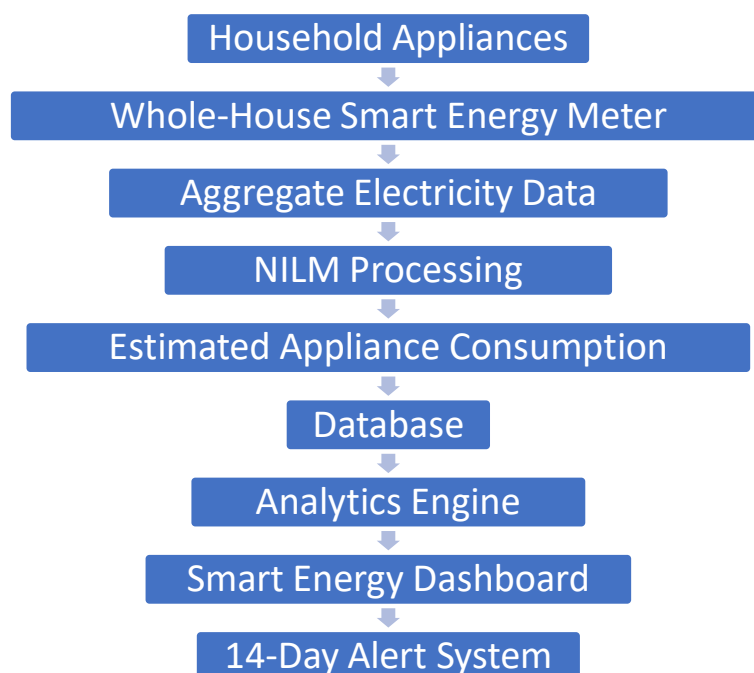
The main objectives of the proposed system are to:

- Collect aggregate household electricity measurements at regular time intervals.
- Use Non-Intrusive Load Monitoring (NILM) to estimate individual appliance energy consumption.
- Store and display daily and monthly appliance-wise consumption.
- Compare current consumption with historical patterns.
- Analyse consumption over 14-day periods and generate alerts when usage is unusually high.
- Identify the appliances and days mainly responsible for excessive consumption.
- Help users take appropriate actions to reduce unnecessary energy use.

System Working

A whole-house smart energy meter records the combined electricity consumption of the household. The collected time-series data is provided to the NILM system. NILM analyses changes and patterns in the aggregate electricity signal and estimates the contribution of individual appliances such as fans, refrigerators, lights, televisions and air conditioners.

The basic workflow is:



Data Processing and Storage

The NILM system produces estimated power consumption for individual appliances. These values are converted into energy consumption using:

$$E = P \times t$$

where

E is energy

P is power

t is time.

The calculated values are stored along with the appliance name, date and time. This allows the system to generate daily, monthly and historical consumption records.

Dashboard and Alert System

The dashboard displays daily appliance-wise consumption, total consumption, monthly comparisons, graphs and historical trends.

Daily data can be viewed continuously, but alerts are evaluated over a 14-day period. This prevents normal day-to-day variations from producing unnecessary warnings. If consumption is significantly higher than the expected level, the system generates an appropriate alert such as Normal, Moderate, High or Critical.

The alert also identifies the main contributing appliance and the days with the highest consumption, making the information more useful for taking corrective action.

Proposed System Output

For example, the system may display:

High Consumption Alert: 21% above expected

Main contributor: Air conditioner

Highest-consumption days: Day 4, Day 7 and Day 11

Thus, the proposed solution converts aggregate electricity measurements into understandable appliance-level information and actionable energy insights.

Feasibility Analysis

The proposed NILM-based energy monitoring system is feasible because it combines existing technologies such as smart energy meters, NILM algorithms, databases and web-based dashboards.

Technical Feasibility

The system requires aggregate household electricity measurements collected at regular time intervals. This data can be processed by a NILM model to estimate individual appliance consumption. During the initial development stage, publicly available NILM datasets can be used to test the system before connecting it to real household data.

Economic Feasibility

Installing an individual energy-monitoring device for every appliance can increase the overall cost, especially in households with many appliances. The proposed approach reduces this requirement by using whole-house measurement and software-based disaggregation. Therefore, it can be more scalable than monitoring every appliance separately.

Operational Feasibility

The system is designed to operate automatically. After the initial setup, electricity data can be collected, processed and stored without manual recording. Users can access the dashboard to view consumption and receive alerts.

Limitations

NILM provides estimated appliance consumption rather than direct measurements. Its accuracy may be affected by simultaneous operation of appliances, similar electrical signatures, data quality and sampling frequency. Therefore, the system should present appliance-level values as estimates.

Expected Environmental Benefits

The proposed system can help reduce unnecessary electricity consumption by making hidden appliance-level energy usage visible to users.

The major expected benefits include:

- **Reduction of energy waste:** Users can identify appliances consuming more energy than expected.
- **Improved energy awareness:** Appliance-wise information helps users understand where electricity is being used.

- **Better energy management:** Daily and monthly records allow users to observe changes in consumption.
- **Targeted corrective action:** Alerts identify the appliances mainly responsible for excessive consumption.
- **Long-term improvement:** Historical data allows users to monitor whether their energy consumption is improving over time.

The system therefore encourages users to move from simply checking their electricity bill to actively monitoring and managing their energy consumption. This can contribute to more responsible electricity use and support the objectives of sustainable energy management.

Justification of the Proposed Solution

Conventional electricity meters mainly provide the total electricity consumed by a household. They do not normally explain how much of that consumption comes from individual appliances. Installing a separate energy-monitoring device for every appliance can provide this information, but it may be expensive and inconvenient, particularly for fixed appliances such as ceiling fans and lights.

The proposed system uses **Non-Intrusive Load Monitoring (NILM)** to overcome this limitation. A whole-house smart meter provides the aggregate electricity signal, and NILM processes this signal to estimate appliance-level consumption.

The proposed system is further improved by adding a historical analysis and intelligent alert layer. It does not simply show the estimated energy values. It compares current consumption with an expected historical baseline and analyses the result over a 14-day period.

For example, instead of simply displaying:

“High electricity consumption detected.”

the system can provide:

“Consumption is 21% above the expected level. The air conditioner is the main contributor, with the highest consumption occurring on Days 4, 7 and 11.”

Thus, the system provides information about how much was consumed, which appliance caused the increase and when the increase occurred. This makes the solution more useful for practical energy management.

Conclusion

The proposed **NILM-Based Smart Energy Monitoring and Intelligent Alert System** provides a practical approach to monitoring and reducing high household electricity consumption. Conventional electricity meters mainly provide the total energy consumed by a household and do not clearly show which individual appliances are responsible for that consumption. The proposed system addresses this limitation by using **Non-Intrusive Load Monitoring (NILM)** to estimate appliance-level energy consumption from aggregate electricity measurements collected through a whole-house smart meter.

The estimated appliance data is stored in a database and presented through a user-friendly dashboard. The dashboard allows users to view daily consumption, monthly summaries, appliance-wise comparisons and long-term consumption trends. This makes it easier to understand household energy usage and identify appliances that consume a significant amount of electricity.

A major feature of the proposed system is its **14-day intelligent alert mechanism**. Instead of generating alerts for every daily variation, the system compares consumption over a 14-day period with an expected value based on historical data. If consumption is significantly higher than expected, the system generates an appropriate alert level. It can also identify the main appliances contributing to the increase and the specific days on which their consumption was highest.

Therefore, the system does more than simply display electricity consumption. It provides useful information about how much energy is being consumed, which appliances are responsible for high consumption and when the increase occurs. This allows users to take more targeted corrective actions and develop better energy-saving habits.

The initial system can be developed using publicly available NILM datasets, allowing the software and analysis methods to be tested before using real household data. In the future, the system can be enhanced with advanced machine learning, seasonal comparison, electricity-bill prediction, carbon-footprint estimation and mobile notifications.

Overall, the proposed system combines smart energy measurement, NILM, data analysis, visualization and intelligent alerts to support efficient electricity management. By increasing awareness and helping users identify unnecessary energy consumption, it can contribute to energy conservation, responsible electricity use and sustainable household energy management.

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